

Sound and uncertainty: why perfect knowledge of audio is physically impossible

The uncertainty principle is not merely a quantum curiosity — it is the single deepest constraint governing every system that produces, transmits, analyzes, or perceives sound. From the Fourier mathematics of signal processing to the quantum behavior of phonons in crystal lattices, from the cochlea's mechanical filtering to the design of MP3 codecs and bat echolocation, one rule persists: **you cannot simultaneously know exactly when a sound occurs and exactly what frequency it contains**. This constraint is not a technological limitation waiting to be overcome. It is a mathematical property of waves themselves, [Subsurfwiki](#) [JMFT](#) and it shapes everything from how pianos are tuned to how ultrasound images are formed. The deep unity here is striking: the same inequality — $\sigma_t \times \sigma_f \geq 1/(4\pi)$ — governs Dennis Gabor's 1946 theory of communication, Werner Heisenberg's quantum mechanics, and the human auditory cortex's processing strategy.

The mathematics behind the Gabor limit

The Fourier uncertainty principle states that a signal's spread in time (σ_t) and its spread in frequency (σ_f) are bound by the inequality $\sigma_t \times \sigma_f \geq 1/(4\pi)$. [Stony Brook University](#) In practical engineering terms, the duration-bandwidth product satisfies $\Delta f \times \Delta t \geq 1/2$.

[Springer](#) This is not an approximation or a measurement artifact — it is a theorem of Fourier analysis. A function and its Fourier transform cannot both have compact support, which means a signal cannot be both time-limited and band-limited simultaneously. [LinkedIn](#) A signal cannot plot as a single point on the time-frequency plane; [Agile](#) it must occupy a minimum area of $1/(4\pi)$, often called a Heisenberg box.

Only one signal shape achieves exact equality: the Gaussian pulse, [Wikipedia](#) which Gabor called an “elementary signal” or Gabor atom. The Gabor atom — a Gaussian-windowed complex sinusoid $g(t) = e^{(-\pi(t-\tau)^2)} \times e^{(j2\pi\omega t)}$ — tiles the time-frequency plane into information cells of minimum area. Every other signal shape occupies a larger region. This has immediate practical consequences. A pure 440 Hz tone played for one second has excellent frequency definition but poor time localization — it extends over 44,100 samples. A sharp drumstick click has superb temporal precision but its spectrum fans across the entire audible range. The briefer the sound, the broader its observed spectrum, [Springer](#) and no amount of computational ingenuity changes this.

The mathematical structure is identical to Heisenberg's quantum uncertainty principle (σ_x

$\times \sigma_p \geq \hbar/2$), because both arise from the same Fourier transform property: position and momentum are conjugate variables [Wikipedia](#) related by the de Broglie relation, [Fiveable](#) just as time and frequency are conjugate variables in signal analysis. [unsw](#) [Wikipedia](#) The critical difference is scale. Heisenberg's constant $\hbar/2 \approx 5.3 \times 10^{-35}$ J·s is negligible at human scales, [unsw](#) [AMS](#) while Gabor's limit of $1/(4\pi) \approx 0.08$ is directly observable every time an audio engineer adjusts an FFT window size.

How spectrograms and the STFT navigate the trade-off

The Short-Time Fourier Transform is the workhorse of audio analysis, [Springer](#) and its behavior is a textbook demonstration of the uncertainty principle. The STFT chops a signal into overlapping windows, computes the Fourier transform of each, and stacks the results into a spectrogram. [ResearchGate](#) The window length directly controls the trade-off:

[Wikipedia](#) **frequency resolution equals the sample rate divided by the window length** [Spectraplus](#) (f_s/N), while time resolution equals the window duration (N/f_s).

At 44.1 kHz sampling, a 256-sample window (~5.8 ms) gives ~172 Hz frequency resolution — adequate for tracking transients like drum hits but unable to distinguish notes separated by less than three semitones in the bass register. [Audacity](#) A 4096-sample window (~93 ms) resolves frequencies to ~10.8 Hz — sufficient to separate adjacent piano keys — but smears temporal events over nearly a tenth of a second. [IRCAM](#) Speech analysis conventionally uses 1024-sample windows (~23 ms) [Csound](#) with Hamming weighting, striking a balance that captures one phoneme per frame while including at least two cycles of the lowest voice fundamental (~100 Hz). [DSP Related](#)

Window functions further shape this trade-off. The rectangular window delivers the narrowest mainlobe (best frequency resolution) but produces sidelobes only **-13 dB** down, [Stanford CCRMA](#) causing severe spectral leakage. [DSP Related](#) The Hann window doubles the mainlobe width but pushes sidelobes to -31 dB. [Stanford CCRMA](#) The Gaussian window uniquely achieves the theoretical minimum uncertainty product, making the Gabor transform (STFT with Gaussian window) optimally localized in both domains. [Fiveable](#) The Kaiser window offers a tunable parameter β to slide continuously along the trade-off curve. [Stanford CCRMA](#) In practice, most audio applications use the Hann window as a pragmatic compromise, [Wray Castle](#) while the MLT Sine window (with sidelobes at -24 dB) is the standard in MP3 and AAC codecs. [Stanford CCRMA](#)

Wavelets redistribute uncertainty without defeating it

The continuous wavelet transform was developed specifically to address the STFT's fixed-resolution limitation. [Amrita](#) Rather than analyzing every frequency with the same window length, wavelets scale the analysis window with frequency: narrow windows at high

frequencies (good time resolution, coarse frequency resolution) and wide windows at low frequencies (good frequency resolution, coarse time resolution). [University of Central Florida](#)

The result is a tiling of the time-frequency plane with Heisenberg boxes that change aspect ratio — tall and thin at high frequencies, short and wide at low frequencies — while maintaining the same minimum area.

This multi-resolution structure matches both natural signals and human perception remarkably well. High-frequency transients (cymbal attacks, consonant onsets) are brief and benefit from fine temporal resolution. Low-frequency content (bass notes, vowel formants) is sustained and benefits from fine frequency resolution.

[University of Central Florida](#) The Morlet wavelet, essentially a Gabor atom at each scale, achieves the minimum uncertainty area at every frequency. A recent advance called **superlets** (Moca et al., 2021, *Nature Communications*) combines sets of wavelets with geometrically increasing bandwidth constraints, achieving what the authors call “time-frequency super-resolution” [Nature](#) — though this still operates within the fundamental uncertainty bound, optimizing the trade-off rather than breaking it.

The Wigner-Ville distribution represents a different approach: a bilinear time-frequency representation that achieves maximum resolution for single-component signals. [Wikipedia](#) For a lone linear chirp, the WVD concentrates energy perfectly along the instantaneous frequency line — a result no linear method can match. The catch is severe: for multi-component signals, the WVD produces **$N(N-1)/2$ cross-terms** that oscillate wildly and can dominate the representation. [Fiveable](#) Cohen’s class of distributions (smoothed pseudo-WVD, Choi-Williams, Born-Jordan) applies various smoothing kernels to suppress cross-terms at the inevitable cost of resolution [Fiveable](#) — another manifestation of the same fundamental constraint.

Quantum mechanics enters through the phonon

When sound propagates through a solid, the classical wave description gives way to quantum mechanics at sufficiently small scales. The vibrational modes of a crystal lattice, when quantized, yield quasiparticles called **phonons** — the acoustic analog of photons.

[Wikipedia](#) [ACS Publications](#) A phonon with angular frequency ω carries energy $E = \hbar\omega$ and crystal momentum $p = \hbar k$. [University of Warwick](#) Like photons, phonons are bosons [Quantiki](#) (spin-0), obey Bose-Einstein statistics, and can occupy the same quantum state in unlimited numbers.

The quantization proceeds by decomposing the lattice Hamiltonian into independent harmonic oscillators via Fourier transformation to normal-mode coordinates, then promoting classical amplitudes to creation and annihilation operators. [ScienceDirect](#) The resulting quantum Hamiltonian is $H = \sum \hbar\omega_k (a_k^\dagger a_k + 1/2)$. [Pan-training](#) [Wikipedia](#) The $1/2$ term —

the zero-point energy — is a direct consequence of Heisenberg's uncertainty principle: confining an atom to a lattice site while demanding zero momentum would violate $\Delta x \Delta p \geq \hbar/2$. [Wikipedia](#) Even at absolute zero, atoms vibrate with root-mean-square displacements typically **3–5% of the interatomic spacing**. This zero-point motion has dramatic physical consequences: it prevents helium from solidifying at atmospheric pressure down to 0 K, [Profmattstrassler](#) and it was directly detected in perovskite nanocrystals through up-converted photoluminescence in a 2025 experiment by Duan et al. [APS Physics](#)

The energy-time uncertainty relation $\Delta E \Delta t \geq \hbar/2$ governs phonon lifetimes and spectral linewidths. A phonon with lifetime τ has an energy uncertainty $\Delta E \sim \hbar/\tau$, producing a Lorentzian spectral line [UCSD Quantum Mechanics](#) of width $\Gamma = \hbar/\tau$. Neutron spin-echo measurements in clathrate compounds have measured acoustic phonon linewidths of ~ 100 μeV , corresponding to lifetimes of **15–33 picoseconds**. [PubMed Central](#) The number-phase uncertainty relation $\Delta n \Delta \phi \geq 1/2$ constrains phonon states analogously to photon quantum optics: [Optics & Photonics News](#) a Fock state $|n\rangle$ with definite phonon number has completely indeterminate phase, while a coherent state $|\alpha\rangle$ (the closest analog to a classical sound wave) distributes uncertainty equally between amplitude and phase quadratures. [Wikipedia](#)

From quantum to classical: why everyday sound has no quantum character

A macroscopic sound wave contains an astronomically large number of phonons, placing it firmly in the classical regime. A 1 kHz tone at 60 dB in a solid involves roughly **10^{24} phonons per second per square meter** — the relative quantum uncertainty of $\sim 1/\sqrt{N} \approx 10^{-12}$ is utterly negligible. Even ultrasound at 1 GHz requires $\sim 10^{15}$ phonons per picowatt. Quantum effects emerge only at GHz frequencies combined with millikelvin temperatures, where k_{BT} approaches $\hbar\omega$. [Photonicsonline](#)

This regime has become experimentally accessible through quantum acoustics, an emerging field coupling phonons to superconducting qubits. In 2017, Chu et al. at Yale demonstrated strong coupling between a transmon qubit and a bulk acoustic wave resonator, creating and controlling single-phonon Fock states at GHz frequencies. [Science](#) By 2023, Bild et al. had created Schrödinger cat states of a **16-microgram mechanical oscillator** — macroscopic quantum superpositions of sound. [PubMed](#) [Nature](#) Surface acoustic wave cavities coupled to qubits exploit a key advantage: acoustic waves propagate 10^5 times slower than electromagnetic waves, enabling compact on-chip quantum memories. [Nature](#) These experiments probe the deep mathematical parallel between the Schrödinger equation for lattice dynamics and Newton's classical equations — both yield identical normal-mode structures, but the quantum version adds discreteness, zero-point motion, and entanglement. [ScienceDirect](#)

The particle-wave transition of phonons mirrors the classical-quantum boundary. At low temperatures ($T \sim 10$ K in typical solids), the thermal phonon wavelength reaches ~ 24 nm — far exceeding the lattice constant — and transport becomes ballistic and coherent.

[IntechOpen](#) At room temperature, the dominant phonon wavelength shrinks to ~ 0.8 nm (comparable to interatomic spacing), scattering becomes frequent, and transport is diffusive. Anderson localization of phonons — the acoustic analog of electron localization in disordered potentials — has been observed in GaAs/AlAs superlattices with ErAs nanodots, where thermal conductivity first increases then decreases with increasing superlattice periods, a hallmark of wave interference-driven localization. [PubMed Central](#) [ResearchGate](#)

The cochlea as nature's wavelet analyzer

The human cochlea performs a real-time spectral decomposition that is strikingly similar to a wavelet transform, and its design embodies the uncertainty trade-off. The basilar membrane acts as a bank of approximately **3,500 overlapping bandpass filters**, each tuned to a characteristic frequency. High frequencies excite the cochlear base (near the oval window); low frequencies excite the apex. [NYU](#) Critically, the filter bandwidths are not uniform: they follow the Equivalent Rectangular Bandwidth scale, where $ERB \approx 24.7(0.00437f + 1)$ Hz. [Purdue University](#) At 1 kHz, the ERB is ~ 133 Hz; at 3 kHz, it widens to ~ 349 Hz. [Purdue University](#) This means high-frequency channels have wider bandwidths (better time resolution, coarser frequency resolution) while low-frequency channels are narrower (better frequency resolution, coarser time resolution) — precisely the multi-resolution structure of a wavelet transform.

The cochlear amplifier, powered by outer hair cells expressing the motor protein prestin, provides **40–60 dB of active gain** at low input levels, sharpening the filters dramatically. The quality factor Q_{ERB} in humans reaches ~ 12 – 15 at 1 kHz, making human cochlear tuning **2–3 times sharper than in common laboratory animals** like cats or guinea pigs. The impulse response duration of a cochlear filter scales as approximately $37/F$ milliseconds (F in kHz): ~ 37 ms at 1 kHz, ~ 9 ms at 4 kHz. Longer ringing at low frequencies means better frequency resolution but worse temporal resolution — the uncertainty principle written directly into the cochlea's mechanical properties.

Human hearing both obeys and seemingly transcends the Gabor limit

Frequency discrimination in human hearing follows Weber's law: the just-noticeable difference (JND) for frequency is approximately **0.2–0.5%** of the test frequency across the sensitive range of 500–4000 Hz. [ScienceDirect](#) [Lumen Learning](#) At 1 kHz and 60 dB SPL, this corresponds to ~ 3 Hz — meaning listeners can distinguish 1000 Hz from 1003 Hz given

sufficient duration. Gap detection thresholds for broadband noise reach **2–3 ms** in normal-hearing adults, [PubMed](#) establishing the ear's temporal acuity. Forward masking persists for 100–200 ms after a masker, while backward masking extends only 10–20 ms — a critical asymmetry that audio codecs exploit.

The minimum duration for pitch perception is directly constrained by the uncertainty principle. [unsw +2](#) Reliable pitch identification requires approximately **10–20 cycles** of a pure tone: ~12 ms at 1 kHz, ~50 ms at 100 Hz, and several hundred milliseconds for the lowest piano note (A0 = 27.5 Hz). Below about 2–4 cycles, pitch perception degrades to chance when phase cues are eliminated. Hsieh and Saberi [Springer](#) (2016) demonstrated that earlier claims of sub-cycle pitch extraction were likely based on timbre (spectral shape) cues rather than genuine pitch perception. [Springer](#) The product $\Delta f \times \text{duration}$ remains roughly constant for short tones, consistent with the Gabor limit.

A remarkable 2013 finding by Oppenheim and Magnasco challenged conventional wisdom: human subjects could simultaneously discriminate frequency and timing with a product $\delta t \times \delta f$ that **exceeded the Fourier uncertainty limit by a factor of ~10 in absolute terms**.

[Phys.org +2](#) The gains came primarily through extraordinary timing acuity. The authors argued that the cochlea's nonlinear processing is essential, since linear systems provably cannot exceed the uncertainty limit. [Phys.org](#) While a 2015 comment questioned the experimental design, [ResearchGate](#) the finding underscores that the biological auditory system employs sophisticated nonlinear algorithms — including active amplification, compression, and neural sharpening — that optimize performance within (and perhaps beyond, in practical terms) the fundamental constraint. The dual coding strategy of the auditory nerve — place coding (tonotopic) above ~4 kHz and temporal phase-locking coding below ~3 kHz, with both contributing in the transition zone [NYU +2](#) — reflects the system's strategy for extracting maximum information from an uncertainty-constrained input. [Journal of Neuroscience](#)

Audio compression exploits uncertainty as a feature

Lossy audio codecs like MP3 and AAC are engineered entirely around the uncertainty principle. [AmpVortex](#) The Modified Discrete Cosine Transform partitions audio into overlapping blocks: **2048 samples (~46 ms) for AAC long blocks** yielding 21.5 Hz frequency resolution, or **256 samples (~5.8 ms) for short blocks** when the psychoacoustic model detects a transient. [ScienceDirect](#) Long blocks efficiently encode tonal content by exploiting spectral stationarity; short blocks prevent pre-echo on transients by containing quantization noise within the attack window. [ScienceDirect](#)

Pre-echo is the most audible artifact of the uncertainty principle in codec design. [Wikipedia](#) When a sharp transient occurs near the end of a long transform block preceded by silence,

the transform spreads the transient's energy across all frequency bins. [ScienceDirect](#) Quantization noise on those bins, once inverse-transformed, fills the entire block — including the silent region before the transient. [O'Reilly](#) Forward masking (50–200 ms) renders post-echo inaudible, but backward pre-masking is effective for only **~2–5 ms**, making pre-echo in a 46 ms block clearly audible. [Tonestack](#) At 256 kbps VBR, MP3 smears a sub-2 ms transient over ~35 ms. [Wikipedia](#) [Sound On Sound](#) AAC's adaptive block switching, temporal noise shaping, and pure MDCT architecture (eliminating MP3's hybrid filterbank artifacts) achieve equivalent quality at roughly 30% lower bitrates [TRTC](#) — largely by managing the uncertainty trade-off more intelligently.

The theoretical floor for transparent compression — perceptual entropy — sits at approximately **2.1 bits per sample** for CD-quality audio, compared to 16 bits uncompressed. [Mp3-tech](#) This nine-fold reduction reflects how much information the ear actually discards through masking, a process fundamentally tied to the cochlea's finite time-frequency resolution cells.

Musical instruments occupy different regions of the uncertainty plane

Different instrument families naturally inhabit different positions on the time-frequency trade-off curve, and musicians intuitively navigate this constraint. **Percussion instruments prioritize temporal precision:** a snare drum hit lasting 10–50 ms produces broadband energy from hundreds of Hz to above 10 kHz, with inharmonic membrane modes (the second mode of a circular membrane sits at 1.59× the fundamental, not 2×) that prevent clear pitch sensation. [Physics LibreTexts](#) **Sustained instruments prioritize frequency precision:** a violin or organ sustains notes for seconds, allowing the ear's pitch mechanisms to operate with maximum accuracy. Musicians performing long sustained chords instinctively tune more carefully than during fast percussive passages — a behavioral response to the uncertainty principle. [unsw](#)

The piano elegantly bridges both regimes. The hammer's contact time is **~2 ms in the treble and ~4 ms in the bass**, producing a broadband attack transient that carries timing information. After the hammer rebounds, the strings vibrate with near-harmonic partials that define pitch. But piano strings have finite stiffness, creating inharmonicity: the n th partial follows $f_n = n \cdot f_1 \cdot \sqrt{1 + Bn^2}$, where B depends on string geometry. [ResearchGate](#) In bass notes of small uprights, upper partials can be shifted several percent sharp of their harmonic positions. [Wikipedia](#) This produces the famous Railsback curve — pianos are tuned with stretched octaves (highs sharper, lows flatter than equal temperament) so that partials of different notes align perceptually. [Wikipedia](#)

Sonar and echolocation reveal the trade-off as engineering problem

Active sonar confronts the uncertainty principle as a range-versus-velocity dilemma. A short pulse provides excellent range resolution ($\Delta r = c/2B$, where B is bandwidth) but poor velocity resolution; a long pulse reverses the trade-off. Chirp signals — frequency sweeps with time-bandwidth products of 100 to 10,000+ — solve this by compressing via matched filtering into pulses far shorter than the original transmission while preserving the energy of a long emission. [Dspguide](#) [Wikipedia](#) A CHIRP fishfinder transmitting 130–210 kHz puts 10–50× more energy into the water than conventional sonar despite lower peak power, achieving simultaneously fine range and Doppler resolution. [West Marine](#)

Medical ultrasound faces the same constraint: axial resolution equals half the spatial pulse length, so shorter pulses (fewer cycles, higher frequency) resolve finer structures but attenuate faster with depth. [Dccam](#) At **2 MHz**, axial resolution is ~1 mm with 15–20 cm penetration. [NYSORA](#) At **10 MHz**, resolution improves to ~0.15 mm but penetration drops to 1–3 cm. Pulsed-wave Doppler introduces yet another uncertainty dimension: higher pulse repetition frequencies extend the unambiguous velocity range but create range ambiguity, while continuous-wave Doppler sacrifices all range information for unlimited velocity measurement. [Echocardiografie](#)

Bats have evolved perhaps the most elegant biological solutions to the uncertainty trade-off. FM (frequency-modulated) bats like vespertilionids emit broadband downward sweeps [Jerseybatgroup](#) [Wikipedia](#) (110→20 kHz in <5 ms), [PNAS](#) achieving excellent range resolution through pulse compression but limited detection range due to energy dilution across frequencies. [Wikipedia](#) CF (constant-frequency) bats like rhinolophids emit long pure tones [Wikipedia](#) (20–60 ms at ~80 kHz), concentrating all energy at one frequency for maximum detection range and exquisite Doppler sensitivity to prey wing flutter — at the cost of spatial resolution. [Jerseybatgroup](#) [Wikipedia](#) CF-FM hybrid species combine both strategies: the CF component detects motion while FM bookends provide range information. [Science News Today](#) CF bats even perform active **Doppler shift compensation**, lowering their call frequency during approach to keep echoes within a narrow cochlear “acoustic fovea” — a biological frequency-tracking servo system. [Wikipedia](#)

Conclusion

The uncertainty principle in sound is not one phenomenon but a single mathematical truth manifesting across every scale and domain of acoustics. The Fourier inequality $\sigma_t \times \sigma_f \geq 1/(4\pi)$ is the classical signal-processing form; Heisenberg's $\Delta x \Delta p \geq \hbar/2$ is the quantum-mechanical form governing phonon behavior; and the cochlea's mechanical filter bank is

evolution's solution to operating within the same constraint. The deepest insight is that these are not analogies — they are the same mathematics applied at different scales.

[LinkedIn +2](#) The STFT window-size dial that every audio engineer adjusts, the block-switching logic inside every AAC encoder, the basilar membrane's graded stiffness, the bat's choice between FM and CF calls, and the zero-point vibration of atoms at absolute zero all trace back to the impossibility of a signal being simultaneously localized in conjugate domains. Understanding this unity transforms the uncertainty principle from a limitation into a design principle — one that nature exploited in the cochlea long before Gabor formalized it, and one that modern quantum acoustics is now harnessing to build phonon-based quantum computers and transducers operating at the single-quantum level.